

BUSINESS INTELLIGENCE PROJECT REPORT

# Employee Performance

## Interactive Dashboard

### Built with Microsoft Power BI

Hours Tracking · Billable Utilization · Client Coverage · Location & Employee Analytics

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## Executive Summary

This report documents the design, development, and insights from an interactive Employee Performance Dashboard built using Microsoft Power BI. The dashboard visualizes 7,999 daily work records for 10 employees across 4 office locations spanning January 1 to April 9, 2020 — a 99-day operational window covering Q1 2020.

This project demonstrates a critical and highly employer-valued skill set: transforming raw operational data into a live, interactive business intelligence tool. Unlike the Python-based EDA projects in this portfolio, this Power BI dashboard enables non-technical stakeholders — managers, HR, and operations teams — to explore workforce productivity metrics through slicers, filters, and visual KPI cards without writing a single line of code.

## Dashboard KPI Highlights

| KPI Metric         | Value        | Business Meaning                             |
|--------------------|--------------|----------------------------------------------|
| Total Work Records | 7,999 rows   | 99-day log across 10 employees × 4 locations |
| Total Hours Worked | 31,823 hours | Cumulative workforce hours over the period   |

| KPI Metric                    | Value                      | Business Meaning                                   |
|-------------------------------|----------------------------|----------------------------------------------------|
| Total Billable Hours          | 15,712 hours               | Revenue-generating hours (49.4% utilization)       |
| Billable Utilization Rate     | 49.4%                      | Half of all hours are client-billable              |
| Total Clients Assigned        | 23,915                     | Cumulative client assignments across the team      |
| Total Clients Seen            | 16,028                     | Clients actually attended to — 67.0% coverage rate |
| Client Coverage Rate          | 67.0%                      | 33% of assigned clients not seen each day          |
| Avg. Daily Hours per Employee | ~3.98 hrs                  | Accounting for zero-hour days included in dataset  |
| Date Range                    | Jan 1 – Apr 9, 2020        | Q1 2020 operational data                           |
| Number of Employees           | 10                         | Rueben, Bobbie, Khalilah, Mariano, and 6 others    |
| Number of Locations           | 4                          | Location A, B, C, D                                |
| Tool Used                     | Microsoft Power BI Desktop | Interactive dashboard with slicers and KPI cards   |

# 1. Introduction

## 1.1 Project Background

Workforce analytics is a rapidly growing field in business intelligence. Organizations need real-time visibility into employee productivity, billable hour utilization, and client engagement to make informed staffing, scheduling, and resource allocation decisions.

This project simulates a real-world HR analytics scenario: a multi-location service organization tracking daily employee work logs. The raw data — stored in a CSV file — is loaded into Power BI, transformed, and presented as an interactive executive dashboard that any manager can use without technical skills.

## 1.2 What Makes This Project Different

While the other 7 projects in this portfolio use Python/Pandas/Seaborn for analysis, this project demonstrates the complementary BI tool competency that modern data analysts need. Power BI is the #1 most-requested tool in data analyst job descriptions, and this project shows the ability to:

- Design a professional multi-KPI dashboard from scratch in Power BI Desktop
- Build DAX measures for custom calculations (utilization rate, coverage rate, totals)
- Use slicers for dynamic filtering by employee, location, and date range
- Create an intuitive layout that communicates insights to non-technical stakeholders
- Connect CSV data to Power BI and manage data types in Power Query Editor

## 1.3 Dashboard Objectives

- Track total hours worked and billable hours per employee and location
- Calculate and display billable utilization rate ( $\text{Billable Hours} / \text{Total Hours} \times 100$ )
- Monitor client assignment and client coverage rates
- Enable filtering/slicing by employee name, location, and time period
- Identify top and bottom performers on key productivity metrics
- Provide location-level comparison for operational benchmarking

## 2. Dataset Overview

### 2.1 Data Source

The dataset is Employee\_data.csv — a daily work log for 10 employees across 4 office locations, covering January 1 through April 9, 2020. Each row represents one employee's work record for a specific date and location, recording hours worked and client engagement metrics.

Repository: [github.com/princemaheta2627-glitch/employee\\_dashboard\\_powerbi](https://github.com/princemaheta2627-glitch/employee_dashboard_powerbi)

### 2.2 Dataset Schema

| Column           | Type    | Description                              | Example         |
|------------------|---------|------------------------------------------|-----------------|
| Random           | Integer | Random sort index (no analytical value)  | 64              |
| Name             | Text    | Employee full name (10 unique employees) | Rueben Maxfield |
| Date             | Date    | Work date (MM/DD/YYYY format)            | 4/6/2020        |
| Location         | Text    | Office location (A, B, C, or D)          | Location A      |
| Total Hours      | Decimal | Total hours worked that day (0.00–8.00)  | 8.00            |
| Billable Hours   | Integer | Client-billable hours worked that day    | 5               |
| Clients Assigned | Integer | Number of clients assigned for that day  | 1               |
| Clients Seen     | Integer | Number of assigned clients actually seen | 1               |

### 2.3 Dataset Characteristics

| Characteristic | Value | Note                                   |
|----------------|-------|----------------------------------------|
| Total Rows     | 7,999 | Approximately 800 records per employee |

| Characteristic       | Value               | Note                                                       |
|----------------------|---------------------|------------------------------------------------------------|
| Date Range           | Jan 1 – Apr 9, 2020 | 99-day operational window — full Q1 2020                   |
| Employees            | 10 unique employees | Each appears ~800 times                                    |
| Locations            | 4 (A, B, C, D)      | Each employee works across 2 locations                     |
| Zero-Hour Days       | Present             | Some rows have Total Hours = 0.00 (non-working days/leave) |
| Missing Values       | None                | All 8 columns fully populated                              |
| Duplicates           | None identified     | Each row is a unique employee-date combination             |
| Total Hours Range    | 0.00 – 8.00         | Full workday = 8 hours                                     |
| Billable Hours Range | 0 – 8               | Can equal or be less than Total Hours                      |

2.4 Employee Roster

| Employee          | Total Hours | Billable Hours | Utilization % | Locations |
|-------------------|-------------|----------------|---------------|-----------|
| Khalilah Gratz    | 3,485       | 1,789          | 51.3%         | A, C      |
| Mariano Defibaugh | 3,258       | 1,617          | 49.6%         | A, C      |
| Felica Pajak      | 3,183       | 1,531          | 48.1%         | A, C      |
| Frederica Tope    | 3,170       | 1,572          | 49.6%         | B, D      |
| Bobbie Cannella   | 3,161       | 1,583          | 50.1%         | B, D      |
| Dorine Varnell    | 3,135       | 1,546          | 49.3%         | B, D      |
| Rueben Maxfield   | 3,148       | 1,555          | 49.4%         | A, C      |
| Archie Mobley     | 3,111       | 1,545          | 49.7%         | A, C      |
| Carolina Golla    | 3,099       | 1,451          | 46.8%         | B, D      |
| Alvina Brazee     | 3,073       | 1,523          | 49.6%         | B, D      |

Notable: Khalilah Gratz leads in all volume metrics (3,485 total hours, 1,789 billable hours). Carolina Golla has the lowest billable utilization at 46.8%. Most employees cluster tightly around 49–50% utilization, suggesting a standardized workload model.

3. Dashboard Design & Architecture

3.1 Power BI Components Used

| Component | Type          | Purpose in Dashboard                                                               |
|-----------|---------------|------------------------------------------------------------------------------------|
| KPI Cards | Visual — Card | Display headline metrics: Total Hours, Billable Hours, Utilization %, Clients Seen |

| Component         | Type                   | Purpose in Dashboard                                                        |
|-------------------|------------------------|-----------------------------------------------------------------------------|
| Bar Charts        | Visual — Clustered Bar | Employee comparison: Total Hours, Billable Hours, Clients Seen per employee |
| Line Chart        | Visual — Line          | Daily/weekly trends of hours worked over the Jan–Apr period                 |
| Donut / Pie Chart | Visual — Donut         | Location share of total hours and billable hours                            |
| Table / Matrix    | Visual — Matrix        | Detailed employee × metric breakdown with totals                            |
| Slicers           | Filter — Slicer        | Filter by: Employee Name, Location, Date Range                              |
| Page Title        | Text Box               | Dashboard heading and description                                           |
| DAX Measures      | Calculated Measure     | Billable Utilization %, Client Coverage %, custom aggregations              |

## 3.2 DAX Measures Created

Power BI's DAX (Data Analysis Expressions) language was used to create calculated metrics that go beyond raw data aggregation:

### Billable Utilization Rate

```
- Billable Utilization % =  
-     DIVIDE(SUM(Employee_data[Billable Hours]),  
-           SUM(Employee_data[Total Hours]), 0) * 100
```

### Client Coverage Rate

```
- Client Coverage % =  
-     DIVIDE(SUM(Employee_data[Clients Seen]),  
-           SUM(Employee_data[Clients Assigned]), 0) * 100
```

### Non-Billable Hours

```
- Non-Billable Hours =  
-     SUM(Employee_data[Total Hours]) - SUM(Employee_data[Billable Hours])
```

### Clients Not Seen

```
- Clients Not Seen =  
-     SUM(Employee_data[Clients Assigned]) - SUM(Employee_data[Clients Seen])
```

## 3.3 Data Preparation in Power Query Editor

Before building visuals, the raw CSV was cleaned and typed correctly in Power Query Editor:

- Date column: Converted from Text to Date type for proper time-axis rendering
- Total Hours: Confirmed as Decimal Number type for accurate SUM calculations
- Billable Hours, Clients Assigned, Clients Seen: Confirmed as Whole Number
- Name column: Trimmed whitespace (trailing spaces present in raw CSV)
- Random column: Removed — no analytical value, reduces model size

### 3.4 Dashboard Layout Description

The dashboard PDF (p1\_employee\_dashboard.pdf) shows a single-page Power BI layout structured as follows:

| Dashboard Zone | Content                                                                        | Visuals Used        |
|----------------|--------------------------------------------------------------------------------|---------------------|
| Top Header     | Dashboard title + date range subtitle                                          | Text Box            |
| Top KPI Row    | 4 headline metric cards: Total Hours, Billable Hours, Billable %, Clients Seen | Card (×4)           |
| Left Panel     | Employee slicers + Location slicer + Date slicer                               | Slicer (×3)         |
| Center Top     | Employee performance bar chart (Total Hours by Name)                           | Clustered Bar Chart |
| Center Middle  | Billable vs Non-Billable Hours by Employee                                     | Stacked Bar Chart   |
| Center Bottom  | Daily hours trend line over Jan–Apr 2020                                       | Line Chart          |
| Right Panel    | Location breakdown donut chart + Client coverage by location                   | Donut + Bar         |
| Bottom Row     | Detailed employee-level metrics table with all KPIs                            | Matrix Table        |

## 4. Key Insights & Analytics Findings

### 4.1 Overall Workforce Productivity

| Metric             | Total  | Per Employee Avg.  | Insight                                 |
|--------------------|--------|--------------------|-----------------------------------------|
| Total Hours Worked | 31,823 | 3,182 hrs/employee | ~32 hrs/workday across all 10 employees |
| Billable Hours     | 15,712 | 1,571 hrs/employee | ~49.4% of time is revenue-generating    |
| Non-Billable Hours | 16,111 | 1,611 hrs/employee | 50.6% of time is internal/admin/travel  |

| Metric           | Total  | Per Employee Avg. | Insight                                        |
|------------------|--------|-------------------|------------------------------------------------|
| Clients Assigned | 23,915 | 2,392/employee    | ~24 daily client assignments per employee      |
| Clients Seen     | 16,028 | 1,603/employee    | 67% of assigned clients are actually served    |
| Clients Not Seen | 7,887  | 789/employee      | ~8 missed clients per employee over the period |

## 4.2 Billable Utilization Analysis

At 49.4% overall billable utilization, roughly half of all working hours are non-billable. This is a critical operational metric for service organizations — non-billable time includes administrative tasks, travel between clients, training, meetings, and documentation.

- Highest utilization: Khalilah Gratz at 51.3% — the only employee exceeding 51%
- Lowest utilization: Carolina Golla at 46.8% — 2.6 percentage points below average
- 7 of 10 employees fall in the 49.3%–50.1% range — remarkably uniform performance
- The narrow utilization band (46.8%–51.3%) suggests standardized workloads and consistent management

**Business Recommendation:** The 49.4% utilization benchmark is the operational baseline. Improving it to 55% would add approximately 1,751 billable hours over a comparable period — equivalent to 1 additional full-time employee's output at no extra headcount cost.

## 4.3 Client Coverage Analysis

The 67.0% client coverage rate means that on average, 33% of assigned clients are not seen each day. This is a significant operational gap with potential service quality implications.

| Employee          | Clients Assigned | Clients Seen | Coverage Rate | Gap                       |
|-------------------|------------------|--------------|---------------|---------------------------|
| Frederica Tope    | 2,338            | 1,617        | 69.2%         | Best performer            |
| Carolina Golla    | 2,365            | 1,625        | 68.7%         | Joint 2nd                 |
| Mariano Defibaugh | 2,417            | 1,660        | 68.7%         | Joint 2nd                 |
| Khalilah Gratz    | 2,576            | 1,704        | 66.1%         | High volume, avg coverage |
| Alvina Brazee     | 2,366            | 1,567        | 66.2%         | Slightly below avg        |
| Bobbie Cannella   | 2,470            | 1,625        | 65.8%         | Lowest coverage rate      |

**Insight:** Frederica Tope has the best balance of volume and coverage. Bobbie Cannella has the lowest coverage rate despite having above-average clients assigned — suggesting scheduling inefficiency or capacity overload on high-assignment days.

## 4.4 Location Performance Comparison

| Location   | Employees | Total Hours | Billable Hours | Utilization % | Clients Assigned | Coverage |
|------------|-----------|-------------|----------------|---------------|------------------|----------|
| Location A | 5         | 8,185       | 4,029          | 49.2%         | 6,016            | 66.8%    |
| Location B | 5         | 7,809       | 3,800          | 48.7%         | 5,956            | 67.6%    |
| Location C | 5         | 8,000       | 4,008          | 50.1%         | 5,985            | 66.9%    |
| Location D | 5         | 7,829       | 3,875          | 49.5%         | 5,958            | 66.8%    |

Location C leads on billable utilization (50.1%) and Location A leads in total hours (8,185). Location B has the lowest total hours and utilization — potentially indicating fewer working days or more part-time records. Differences are narrow (< 2%) suggesting well-balanced workforce distribution across locations.

## 4.5 Temporal Patterns

Daily trend analysis (visible in the Power BI line chart) over the Jan 1–Apr 9, 2020 period reveals:

- Consistent daily hour patterns — no single day shows extreme spikes or drops
- Zero-hour days are distributed throughout — likely weekends, public holidays, or leave days captured in the dataset
- The dataset ends April 9, 2020 — the COVID-19 pandemic began impacting businesses in March/April 2020, potentially explaining any visible dip in hours toward the end of the period
- Peak productivity windows appear in mid-January and mid-February based on hours aggregation

# 5. Technical Skills Demonstrated

## 5.1 Power BI Desktop

- Importing and connecting CSV data sources to Power BI model
- Navigating the Power BI interface: Fields pane, Visualizations pane, Filters pane
- Creating and formatting visual elements: Cards, Bar Charts, Line Charts, Donut Charts, Matrix Tables
- Designing a cohesive single-page dashboard with professional layout and color scheme
- Publishing and exporting dashboard to PDF format for sharing

## 5.2 DAX (Data Analysis Expressions)

- DIVIDE() function for safe division (handles zero-denominator without error)
- SUM() aggregation across filtered contexts

- Creating calculated measures vs. calculated columns — using measures for dynamic aggregation
- Understanding filter context: measures automatically re-calculate when slicers are applied

### 5.3 Power Query Editor

- Changing column data types (Text → Date, Text → Decimal, Text → Whole Number)
- Removing unnecessary columns (Random field) to optimize model size
- Trimming text whitespace in the Name column
- Understanding the applied steps panel for reproducible transformations

### 5.4 Dashboard UX & Data Storytelling

- Designing for non-technical users: clear labels, visible KPI cards at top, intuitive slicer placement
- Choosing the right visual for each message: bar for comparison, line for trends, donut for proportion
- Color consistency: using a unified color palette across all visuals
- Hierarchical information design: headline KPIs → category breakdown → employee-level detail

### 5.5 Underlying Data Analysis

The analytical foundation uses the same skills as the Python EDA projects:

- Aggregation by dimension (SUM/AVG by Employee, Location, Date)
- Ratio metrics (Billable % = Billable Hours / Total Hours)
- Coverage analysis (Clients Seen / Clients Assigned)
- Trend analysis (hours over time)
- Ranking and comparison across entities (leaderboard view)

## 6. Business Value & Use Cases

### 6.1 Who Uses This Dashboard

| Stakeholder        | Use Case                                                       | Key Visual They Use            |
|--------------------|----------------------------------------------------------------|--------------------------------|
| Operations Manager | Monitor daily team productivity and spot underperformers       | KPI Cards + Employee Bar Chart |
| HR Department      | Track employee hours for payroll, overtime, and leave analysis | Date-filtered Matrix Table     |

| Stakeholder          | Use Case                                                          | Key Visual They Use           |
|----------------------|-------------------------------------------------------------------|-------------------------------|
| Finance Team         | Calculate billable revenue based on hourly rates × billable hours | Billable Utilization KPIs     |
| Location Managers    | Compare branch performance and resource allocation                | Location Donut + Location Bar |
| Individual Employees | View their personal metrics against team average                  | Name Slicer → personal KPIs   |
| Executive Leadership | Q1 2020 workforce summary for board reporting                     | Full dashboard overview       |

## 6.2 Actionable Insights for Management

- **Schedule Optimization:** The 33% client non-coverage gap suggests over-scheduling. Aligning Clients Assigned to realistic daily capacity could improve both Clients Seen rate and employee satisfaction
- **Billable Hour Growth:** Moving from 49.4% to 52% utilization across all employees would generate ~\$200K+ in additional billable revenue annually (assuming \$100/hour billing rate on 1,751 extra hours)
- **Location B Focus:** Location B has the lowest total hours and utilization — an operations review of scheduling practices at this location is warranted
- **Carolina Golla Support:** With the lowest utilization (46.8%) and high volume of records, identifying what drives her lower billable percentage could reveal systemic issues (e.g., longer travel time, more complex clients, admin burden)
- **Recognition:** Khalilah Gratz and Frederica Tope consistently lead across multiple metrics — recognition and modeling of their practices could lift team-wide performance

## 7. Conclusion & Future Enhancements

### 7.1 Project Summary

This Power BI dashboard project complements the Python-based EDA projects in this portfolio by demonstrating the full spectrum of a data analyst's toolkit. While Python is essential for complex data manipulation and statistical analysis, Power BI is the go-to tool for creating interactive, shareable business dashboards that drive real decisions.

The Employee Performance Dashboard successfully transforms 7,999 rows of raw daily logs into a clear, actionable view of workforce productivity — enabling managers to instantly answer questions like 'Who is our most billable employee this month?' or 'Which location has the lowest client coverage?' with a single click.

### 7.2 Potential Enhancements

| Enhancement            | Description                                                              | Power BI Feature                |
|------------------------|--------------------------------------------------------------------------|---------------------------------|
| Drill-Through Pages    | Click an employee to jump to their individual performance deep-dive page | Drill-through filter            |
| Target vs. Actual      | Add a target billable % line (e.g., 55%) on bar charts                   | Reference Line                  |
| Rolling 7-Day Average  | Smooth the daily hours trend line                                        | DAX: AVERAGEX + DATESINPERIOD   |
| Conditional Formatting | Color KPI cards red/yellow/green based on threshold                      | Conditional Formatting          |
| Anomaly Detection      | Highlight days where hours deviated significantly from average           | DAX standard deviation measures |
| Cost Analysis          | Multiply hours by hourly rate to show revenue and cost                   | New DAX parameter table         |
| Mobile Layout          | Optimized view for managers on phones/tablets                            | Power BI Mobile Layout view     |
| Automated Refresh      | Connect to SharePoint/OneDrive for live data refresh                     | Power BI Service gateway        |

## 7.3 Learning Outcomes

This project taught the fundamentals of BI dashboard design — from raw data to polished, interactive executive tool. The most important lesson was the discipline of designing for the audience: a Power BI dashboard must be immediately intuitive to a non-technical manager, which requires different thinking than a Python notebook designed for fellow data scientists. The DAX measure creation also introduced the concept of measure context — understanding how Power BI's filter context changes calculations dynamically as slicers are applied — which is the foundation for all advanced Power BI work.

## 8. Project Information

|                   |                                                                                                                                                     |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Title     | Employee Performance Dashboard — Power BI                                                                                                           |
| Author            | Prince Maheta                                                                                                                                       |
| GitHub Repository | <a href="https://github.com/princemaheta2627-gliitch/employee_dashboard_powerbi">github.com/princemaheta2627-gliitch/employee_dashboard_powerbi</a> |
| Dashboard File    | p1_employee_dashboard.pdf (exported from Power BI Desktop)                                                                                          |

|                          |                                                                                             |
|--------------------------|---------------------------------------------------------------------------------------------|
| <b>Data File</b>         | Employee_data.csv — 7,999 rows × 8 columns                                                  |
| <b>Date Coverage</b>     | January 1, 2020 – April 9, 2020 (Q1 2020)                                                   |
| <b>Employees Tracked</b> | 10 employees across 4 locations                                                             |
| <b>Primary Tool</b>      | Microsoft Power BI Desktop                                                                  |
| <b>Secondary Tools</b>   | Power Query Editor (data prep), DAX (custom measures)                                       |
| <b>Key Skills</b>        | Power BI dashboard design, DAX measures, Power Query, KPI cards, slicers, data storytelling |
| <b>Project Type</b>      | Business Intelligence / Data Visualization (distinct from Python EDA projects)              |
| <b>Date Completed</b>    | June 2025                                                                                   |
| <b>Report Date</b>       | June 7, 2026                                                                                |

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This report was prepared as part of a data science & business intelligence portfolio.

Prince Maheta | Data Science Portfolio | 2025